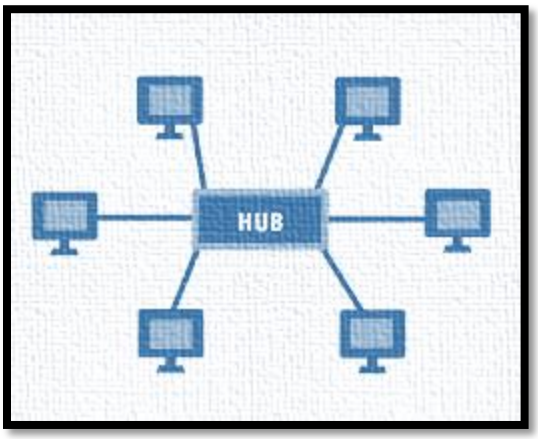
GUJARAT TECHNOLOGICAL UNIVERSITY

Computer Network (3150710) Winter 2021 Paper Solution

Q.1 (a) What is topology? Explain star topology in brief. 03

Topology defines the structure of the network of how all the components are interconnected to each other. A network topology is the physical and logical arrangement of nodes and connections in a network. Nodes usually include devices such as switches, routers and software with switch and router features. Network topologies are often represented as a graph.

Star topology



- Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
- The central computer is known as a server, and the peripheral devices attached to the server are known as clients.
- Coaxial cable or RJ-45 cables are used to connect the computers.
- Hubs or Switches are mainly used as connection devices in a physical star topology.
- Star topology is the most popular topology in network implementation.

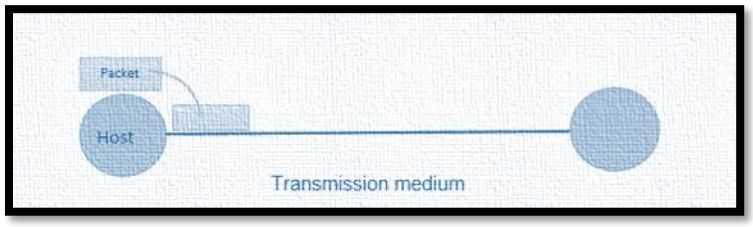
(b) Explain various delay which are occur in data packet transmission. 04

The delays, here, means the time for which the processing of a particular packet takes place. We have the following types of delays in computer networks:

1. Transmission Delay:

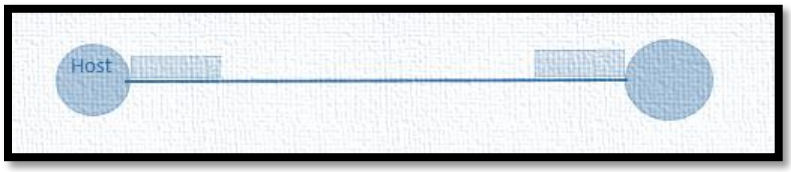
The time taken to transmit a packet from the host to the transmission

medium is called Transmission delay.
For example, if bandwidth is 1 bps (every second 1 bit can be transmitted onto the transmission medium) and data size is 20 bits then what is the transmission delay? If in one second, 1 bit can be transmitted. To transmit 20 bits, 20 seconds would be required.



2. Propagation delay:

After the packet is transmitted to the transmission medium, it has to go through the medium to reach the destination. Hence the time taken by the last bit of the packet to reach the destination is called propagation delay.



Factors affecting propagation delay:

1. **Distance –** It takes more time to reach the destination if the distance of the medium is longer.
2. **Velocity –** If the velocity(speed) of the signal is higher, the packet will be received faster.

3. Queueing delay:

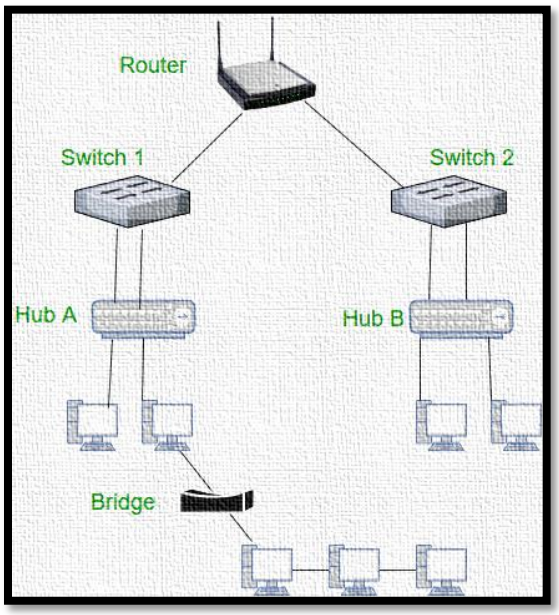
Let the packet is received by the destination, the packet will not be processed by the destination immediately. It has to wait in a queue in something called a buffer. So the amount of time it waits in queue before being processed is called queueing delay.

4. Processing delay:

Now the packet will be taken for the processing which is called processing delay.

Time is taken to process the data packet by the processor that is the time required by intermediate routers to decide where to forward the packet, update TTL, and perform header checksum calculations.

(c) Explain functionality of Repeater, Hub, Bridge, Switch, Router and Gateway. 07



1. Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting if original strength. It is a 2-port device.

2. Hub – A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

3. Bridge – A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

4. Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.

5. Routers – A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.

6. Gateway – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

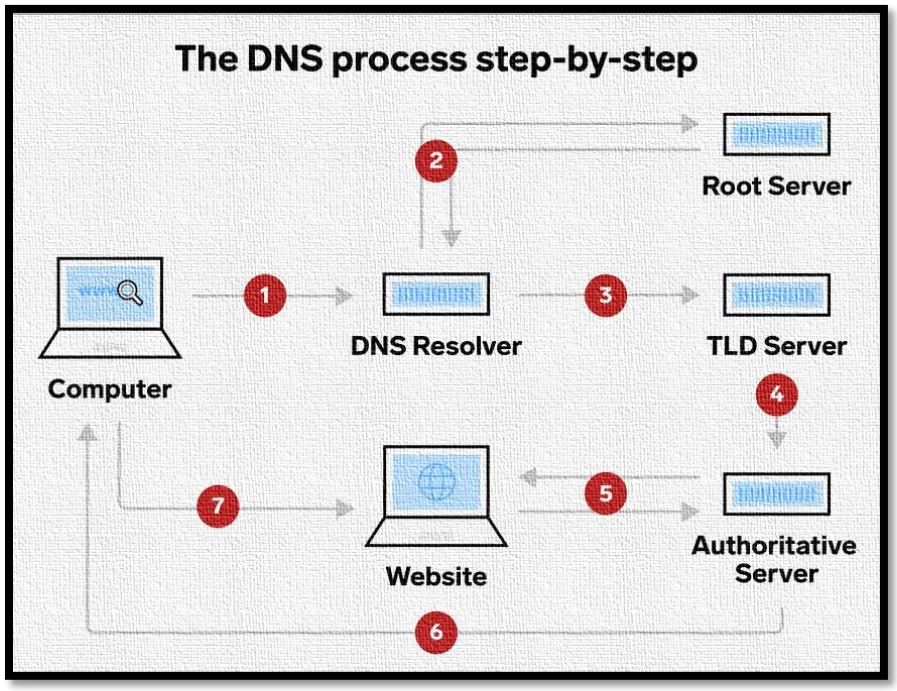
Q.2 (a) Write short note on Domain Name System (DNS). 03

Domain Name System (DNS) is a hierarchical naming system built on a distributed database for computers, services, or any resource connected to the Internet or a private network. Most importantly, it translates human readable domain names into the numerical identifiers associated with networking equipment, enabling devices to be located and connected worldwide. Analogous to a network "phone book," DNS is how a browser can translate a domain name (e.g., "facebook.com") to the actual IP address of the server, which stores the information requested by the browser.

The eight steps in a DNS lookup:

1. A user enters a domain name (e.g., facebook.com) into their browser, and the browser sends the query via their internet service provider isp to a DNS recursive resolver.
2. The DNS recursive resolver, in turn, sends a query to the root DNS nameserver (.).
3. The root server returns to the resolver the address of the top-level domain (i.e., "TLD") DNS root server, which has the needed information for the facebook.com domain. (Examples of a top- level domain tld include

".com", ".net", and ".org," which each TLD having it's own root DNS server.)
4. In turn, the resolver then sends the information request to the Top-Level Domain server (In this case, the ".com" TLD nameserver).
5. The TLD name server responds to the resolver with the targeted IP address of the domain's nameserver. (In this case, the DNS server for "facebook.com".)
6. Next, the DNS recursive resolver sends the query to the domain's DNS server.
7. The domain's DNS server then returns the IP address to the DNS resolver for the requested domain (e.g., "facebook.com").
8. Finally, the DNS resolver returns the IP address of the requested domain to the requesting web browser. The browser sends the HTTPS request to the targeted IP address, and the server with that address returns the webpage, which renders in the user's browser.



(b) What is HTTP? Differentiate its persistent and non-persistent types with request-response behavior of HTTP. 04

Hypertext Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML. It was designed for communication between web browsers and web servers, but it can also be used for other purposes.

Non-persistent and **persistent** are the two types of HTTP connections used to connect the client with the webserver. The non-persistent connection has

connection type 1.0, while the persistent connection has connection type 1.1.

Non-persistent

The non-persistent connection takes a total time of 2RTT + *file transmission time*. It takes the first RTT (round-trip time) to establish the connection between the server and the client. The second RTT is taken to request and return the object. This case stands for a single object transmission.

After the client receives the object in non-persistent, the connection is immediately closed. This is the basic difference between persistent and non-persistent. The persistent connection ensures the transfer of multiple objects over a single connection.

Persistent

A persistent connection takes 1 RTT for the connection and then transfers as many objects, as wanted, over this single connection.

RTT stands for the round-trip time taken for an object request and then its retrieval. In other words, it is the time taken to request the object from the client to the server and then retrieve it from the server back to the client.

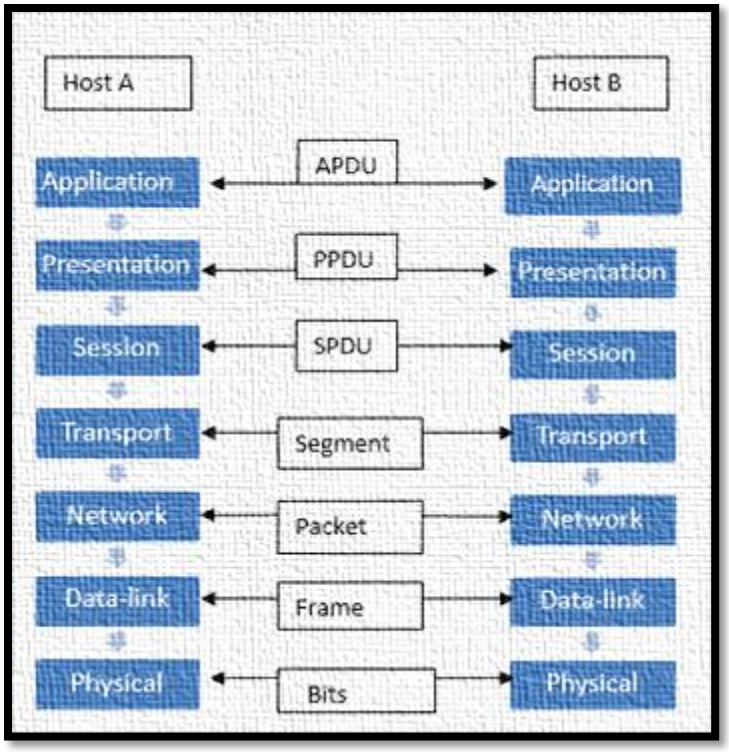
(c) Draw the layered architecture of OSI reference model and write at least two services provided by each layer of the model. 07

OSI is the reference model designed to divide the entire communication operation into multiple processes. This model divides basic communication functions into seven different layers as shown in the figure below. Here two end communication devices are named Application A and Application B. Each layer communicates with their adjacent layer while adding or removing header and trailer.

When the data is being transmitted it goes from bottom to top in the OSI layer architecture. Each layer follows its own function:

1. Physical layer:

- This layer deals with the transfer of bits over a communication channel, for example copper wire pairs, coaxial cable, optical fibre.
- This layer is concerned with the particular choice of system parameters such as voltage level and signal duration, also setup and release of physical connection as well as mechanical aspect such as socket type and number of pins.



2. Data link layer:

- This layer converts bits into group of bits called frames. It also adds additional information related to frame such as address and correction bits and bits to define the boundaries of frames.
- This control is practically important when the transmission link is prone to transmission error. Check bits ensures the detection of error if any happens during transmission also address bits help in the flow control frames.

3. Network Layer:

- This layer combines multiple frames forming a packet while sending the data and when it receives a data it separates frames from the packet.
- This layer provides routing of packet across the communication network and when doing so it is required that the nodes through which packets are passing are always connected to the network. This makes the network layer most complex layer in the OSI model.
- This layer also deals with the congestion that occurs from time to time due to surge in the traffic in the network.

4. Transport Layer:

- The role of this layer is to provide appropriate address information so that the messages can be delivered to the appropriate destination session layer.
- It also performs various functions such as error free transfer of information by the error detection and recovery techniques and sequence and flow control.
- This layer also does the segmentation and reassembly or blocking and unblocking to match the size of the message produced by the session layer to the packet sizes that can be handled by the network layer.

5. Session Layer:

- This layer enhances the reliability of transfer of data over transport layer by providing means of dialogue between two applications. Such as in some communication manners data transfer is half duplex where both applications takes turn in transferring the data. This facility is supported by session layer.
- This application of session layer becomes useful while transferring large files to provide synchronization points from which error recovery can be initiated instead of transferring large files again.

6. Presentation Layer:

- This layer provides presentation of data to the application layer. Different computers use different coding technique to represent different symbols. It is the function of this layer to convert the data into presentable form in the receiving application.
- In other words this layer converts machine dependent information into machine independent and vice versa at the receiving end.

7. Application Layer:

- This layer closet to the user. It provides user services that are frequently required by the application in communication. These services include e-mail, name services, login service, network management etc.

OR (c) Explain DHCP and Email in detail. 07

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to nay device, or node, on a network so they can communicate using IP (Internet Protocol). DHCP automates and centrally manages these configurations. There is no need to

manually assign IP addresses to new devices. Therefore, there is no requirement for any user configuration to connect to a DHCP based network.

DHCP can be implemented on local networks as well as large enterprise networks. DHCP is the default protocol used by the most routers and networking equipment. DHCP is also called RFC (Request for comments) 2131.

Dynamic Host Configuration Protocol (DHCP) is a client/server protocol that automatically provides an Internet Protocol (IP) host with its IP address and other related configuration information such as the subnet mask and default gateway.

Why use DHCP? Every device on a TCP/IP-based network must have a unique unicast IP address to access the network and its resources. Without DHCP, IP addresses for new computers or computers that are moved from one subnet to another must be configured manually; IP addresses for computers that are removed from the network must be manually reclaimed.

E-mail is defined as the transmission of messages on the Internet. It is one of the most commonly used features over communications networks that may contain text, files, images, or other attachments. Generally, it is information that is stored on a computer sent through a network to a specified individual or group of individuals.

Email messages are conveyed through email servers; it uses multiple protocols within the TCP/IP suite. For example, SMTP is a protocol, stands for simple mail transfer protocol and used to send messages whereas other protocols IMAP or POP are used to retrieve messages from a mail server. If you want to login to your mail account, you just need to enter a valid email address, password, and the mail servers used to send and receive messages.

Q.3 (a) Explain Physical Address, IP address, Port Address in brief. 03

1. IP address:

An Internet Protocol address (IP address) is the logical address of our network hardware by which other devices identify it in a network. IP address stands for Internet Protocol address which is a unique number or a numerical representation that uniquely identifies a specific interface on the network. Each device that is connected to internet an IP address is assigned to it for its unique identification.

Addresses in IPv4 are 32-bits long example, 12.244.233.165

And Addresses in IPv6 are 128-bits
example, 2001:0db8:0000:0000:0000:ff00:0042:7879

2. Physical Address: A physical address (also real address, or binary address), is a memory address that is represented in the form of a binary number on the address bus circuitry in order to enable the data bus to access a particular storage cell of main memory, or a register of memory-mapped I/O device.

3. Port Number:
Port number is the part of the addressing information used to identify the senders and receivers of messages in computer networking. Different port numbers are used to determine what protocol incoming traffic should be directed to. Port number identifies a specific process to which an Internet or other network message is to be forwarded when it arrives at a server. Ports are identified for each protocol and it is considered as a communication endpoint.

(b) Compare IPv4 and IPv6. 04

	IPv4	IPv6
Address length	IPv4 is a 32-bit address.	IPv6 is a 128-bit address.
Fields	IPv4 is a numeric address that consists of 4 fields which are separated by dot (.).	IPv6 is an alphanumeric address that consists of 8 fields, which are separated by colon.
Classes	IPv4 has 5 different classes of IP address that includes Class A, Class B, Class C, Class D, and Class E.	IPv6 does not contain classes of IP addresses.
Number of IP address	IPv4 has a limited number of IP addresses.	IPv6 has a large number of IP addresses.
VLSM	It supports VLSM (Virtual Length Subnet Mask). Here, VLSM means that Ipv4 converts IP addresses into a subnet of different sizes.	It does not support VLSM.

Address configuration	It supports manual and DHCP configuration.	It supports manual, DHCP, auto-configuration, and renumbering.
Address space	It generates 4 billion unique addresses	It generates 340 undecillion unique addresses.
End-to-end connection integrity	In IPv4, end-to-end connection integrity is unachievable.	In the case of IPv6, end-to-end connection integrity is achievable.
Security features	In IPv4, security depends on the application. This IP address is not developed in keeping the security feature in mind.	In IPv6, IPSEC is developed for security purposes.
Address representation	In IPv4, the IP address is represented in decimal.	In IPv6, the representation of the IP address in hexadecimal.
Fragmentation	Fragmentation is done by the senders and the forwarding routers.	Fragmentation is done by the senders only.
Packet flow identification	It does not provide any mechanism for packet flow identification.	It uses flow label field in the header for the packet flow identification.
Checksum field	The checksum field is available in IPv4.	The checksum field is not available in IPv6.
Transmission scheme	IPv4 is broadcasting.	On the other hand, IPv6 is multicasting, which provides efficient network operations.
Encryption and Authentication	It does not provide encryption and authentication.	It provides encryption and authentication.

Number of octets	It consists of 4 octets.	It consists of 8 fields, and each field contains 2 octets. Therefore, the total number of octets in IPv6 is 16.

(c) Explain Distance Vector Routing Algorithm. 07

The Distance vector algorithm is iterative, asynchronous and distributed.

- **Distributed:** It is distributed in that each node receives information from one or more of its directly attached neighbors, performs calculation and then distributes the result back to its neighbors.
- **Iterative:** It is iterative in that its process continues until no more information is available to be exchanged between neighbors.
- **Asynchronous:** It does not require that all of its nodes operate in the lock step with each other.

- The Distance vector algorithm is a dynamic algorithm.
- It is mainly used in ARPANET, and RIP.
- Each router maintains a distance table known as **Vector**.

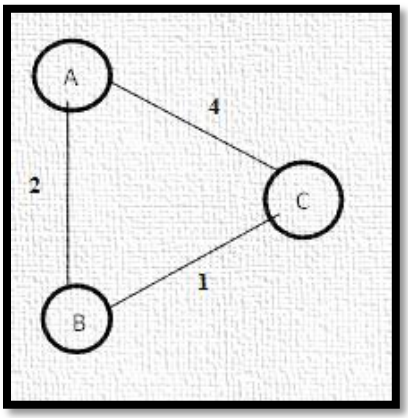
How the DVR Protocol Works

- In DVR, each router maintains a routing table. It contains only one entry for each router. It contains two parts – a preferred outgoing line to use for that destination and an estimate of time (delay). Tables are updated by exchanging the information with the neighbor's nodes.
- Each router knows the delay in reaching its neighbors (Ex – send echo request).
- Routers periodically exchange routing tables with each of their neighbors.
- It compares the delay in its local table with the delay in the neighbor's table and the cost of reaching that neighbor.
- If the path via the neighbor has a lower cost, then the router updates its local table to forward packets to the neighbor.

Example – Distance Vector Router Protocol

In the network shown below, there are three routers, A, B, and C, with the following weights – AB =2, BC =3 and CA =5.

Step 1 – In this DVR network, each router shares its routing table with every neighbor. For example, A will share its routing table with neighbors B and C and neighbors B and C will share their routing table with A.



Form A	A	B	C
A	0	2	3
B			
C			

Form B	A	B	C
A			
B	2	0	1
C			

Form C	A	B	C
A			
B			
C	3	1	0

Step 2 – If the path via a neighbor has a lower cost, then the router updates its local table to forward packets to the neighbor. In this table, the router updates the lower cost for A and C by updating the new weight from 4 to 3 in router A and from 4 to 3 in router C.

Form A	A	B	C
A	0	2	3
B			
C			
Form B	A	B	C
A			
B	2	0	1
C			
Form C	A	B	C
A			
B			
C	3	1	0

Step 3 – The final updated routing table with lower cost distance vector routing protocol for all routers A, B, and C is given below –

Router A

Form A	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

Router B

Form B	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0



Router C

Form C	A	B	C
A	0	2	3
B	2	0	1
C	3	1	0

OR Q.3 (a) Discuss the principles of Reliable Data Transfer. 03

Principles of Reliable Data Transfer

- It's the responsibility of a **reliable data transfer protocol** to implement the service abstraction of providing a reliable data transfer. Which can be hard because the layers bellow it might be unreliable.
- We will use the terminology "packet" rather than transport-layer "segment".
- We will only discuss unidirectional data transfer that is data transfer from the sending to the receiving side. Not bidirectional data transfer (full-duplex)

Refer for more: https://andreasjj.gitbooks.io/ttm4100-kommunikasjon-tjenester-og-nett/content/chapter_3__transport_layer/34-principles-of-reliable-data-transfer.html

(b) Give difference between connection oriented and connectionless services. 04

Comparison Parameter	Connection-oriented Service	Connection Less Service
Related System	It is designed and developed based on the telephone system.	It is service based on the postal system.
Definition	It is used to create an end to end connection between the senders to the receiver before transmitting the data	It is used to transfer the data packets between senders to the receiver without creating any connection.

	over the same or different network.	
Virtual path	It creates a virtual path between the sender and the receiver.	It does not create any virtual connection or path between the sender and the receiver.
Authentication	It requires authentication before transmitting the data packets to the receiver.	It does not require authentication before transferring data packets.
Data Packets Path	All data packets are received in the same order as those sent by the sender.	Not all data packets are received in the same order as those sent by the sender.
Bandwidth Requirement	It requires a higher bandwidth to transfer the data packets.	It requires low bandwidth to transfer the data packets.
Data Reliability	It is a more reliable connection service because it guarantees data packets transfer from one end to the other end with a connection.	It is not a reliable connection service because it does not guarantee the transfer of data packets from one end to another for establishing a connection.
Congestion	There is no congestion as it provides an end-to-end connection between sender and receiver during transmission of data.	There may be congestion due to not providing an end-to-end connection between the source and receiver to transmit of data packets.
Examples	Transmission Control Protocol (TCP) is an example of a connection-oriented service.	User Datagram Protocol (UDP), Internet Protocol (IP), and Internet Control Message Protocol (ICMP) are examples of connectionless service.

(c) What do you mean by congestion and overflow? Explain the slow-start component of the TCP congestion-control algorithm. 07

Congestion: An event where the input rate exceeds the output rate. Frequently used in the specific case of link congestion, where more traffic needs to be transmitted on a link than can be sent. Congestion results in growing the length of the link transmit queue. Sustained congestion is typically indicative of link oversubscription, caused by insufficient network capacity, imperfect traffic engineering, or suboptimal recovery mechanisms.

Overflow: All hardware is finite. While software likes the abstraction that memory is infinite, in any given implementation, it is necessarily finite. When more data is to be stored than can fit in the data store, then you have an overflow condition. Frequently this happens when a transmit queue has filled due to sustained congestion and some packets must be discarded.

Congestion control refers to techniques and mechanisms that can-

- Either prevent congestion before it happens
- Or remove congestion after it has happened

TCP reacts to congestion by reducing the sender window size.

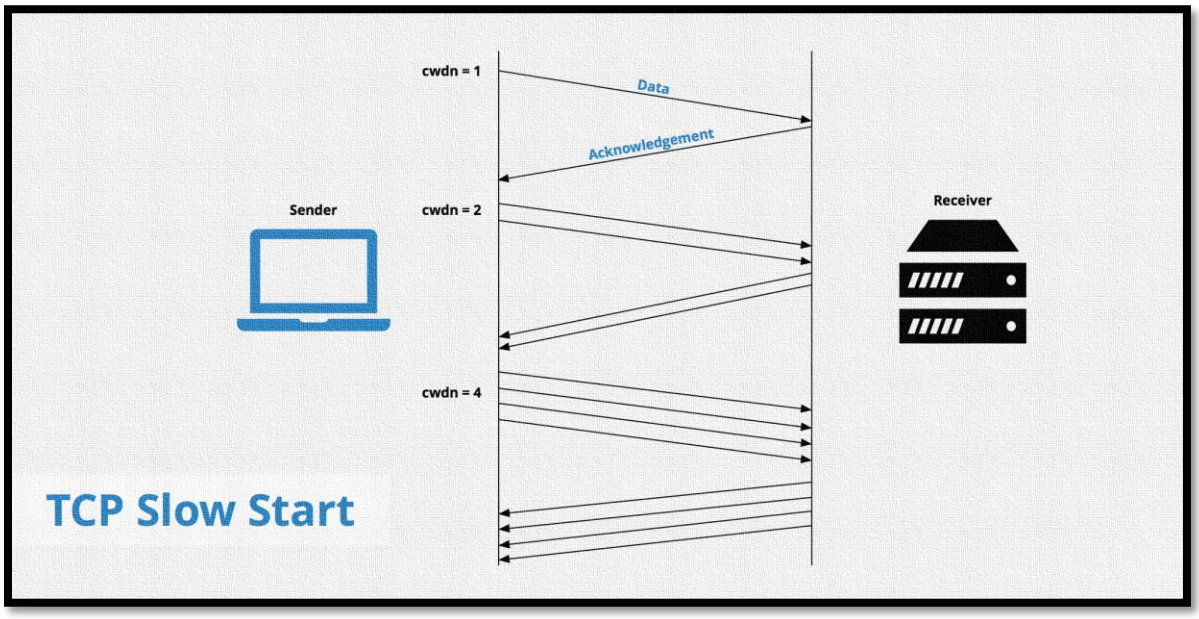
Slow Start Phase:

TCP slow start is part of the congestion control algorithms put in place by TCP to help control the amount of data flowing through to a network. This helps regulate the case where too much data is sent to a network and the network is incapable of processing that amount of data, thus resulting in network congestion. TCP slow start is not the only algorithm put in place to help prevent network congestion, there are four intertwined congestion control mechanisms in total.

When transmission of data from sender to receiver begins in a network, there may be unknown conditions as to what the network can handle. Slow start helps to mitigate the pitfalls of this unknown by implementing the following functionality.

1. A sender begins transmissions to a receiver by slowly probing the network with a packet that contains its initial congestion window (cwnd).
2. The client receives the packet and replies with its maximum buffer size, also known as the receiver's advertised window (rwnd).
3. If the sender receives an acknowledgement from the client, it then doubles the amount of packets to send to the client.

4. Step 3 is repeated until the sender no longer receives acknowledgment from the receiver which means either congestion is detected, or the client's window limit has been reached.



Q.4 (a) Explain packet fragmentation with example. 03

Every packet based network has an MTU (Maximum Transmission Unit) size. The MTU is the size of the largest packet that that network can transmit.

Packets larger than the allowable MTU must be divided into smaller packets or fragments to enable them to traverse the network.

Network	Standard MTU
Ethernet	1500
Token Ring	4096

A Packet Fragmentation Example

If a 2,366 byte packet enters an Ethernet network with a default MTU size, it must be fragmented into two packets.

The first packet will:

- Be 1,500 bytes in length. 20 bytes will be the IP header, 24 bytes will be the TCP header, and 1,456 bytes will be data.

- Have a DF bit equal to 0 to mean "May Fragment" and an MF bit equal to 1 to mean "More Fragments."
- Have a Fragmentation Offset of 0.

The second packet will:

- Be 910 bytes in length. 20 bytes will be the IP header, 24 bytes will be the TCP header, and 866 bytes will be data.
- Have the DF bit equal to 0 to mean "May Fragment" and the MF bit equal to 0 to mean "Last Fragment."
- Have a Fragmentation Offset of 182 (Note: 182 is 1456 divided by 8).

(b) Write a short note on broadcast and multicast routing. 04

Multicast Routing

Multicast routing is special case of broadcast routing with significance difference and challenges. In broadcast routing, packets are sent to all nodes even if they do not want it. But in Multicast routing, the data is sent to only nodes which wants to receive the packets.

Broadcast routing

By default, the broadcast packets are not routed and forwarded by the routers on any network. Routers create broadcast domains. But it can be configured to forward broadcasts in some special cases. A broadcast message is destined to all network devices.

Broadcast routing can be done in two ways (algorithm):

- A router creates a data packet and then sends it to each host one by one. In this case, the router creates multiple copies of single data packet with different destination addresses. All packets are sent as unicast but because they are sent to all, it simulates as if router is broadcasting.

This method consumes lots of bandwidth and router must destination address of each node.

- Secondly, when router receives a packet that is to be broadcasted, it simply floods those packets out of all interfaces. All routers are configured in the same way.

(c) What is IP address? Explain sub netting with example. 07

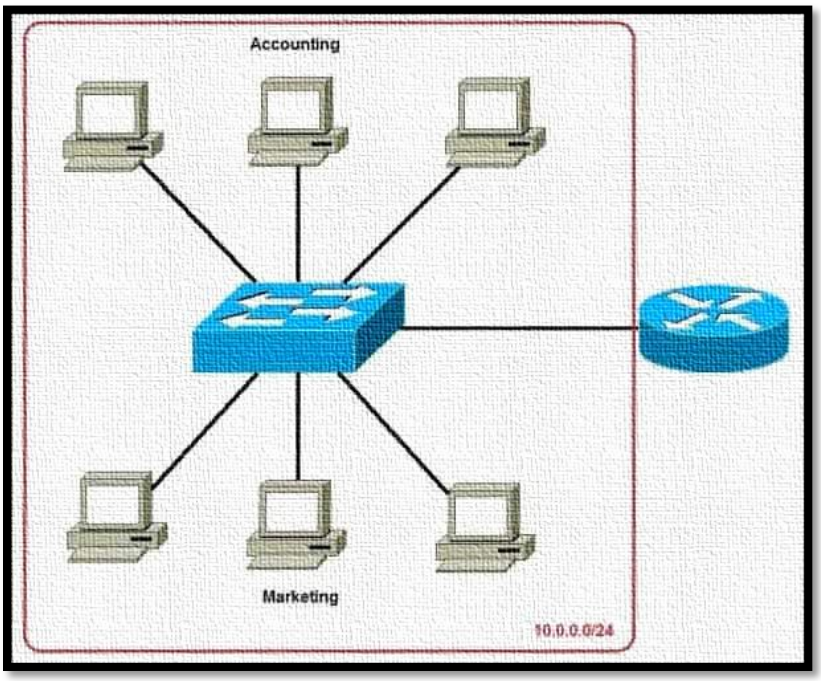
An Internet Protocol address (IP address) is the logical address of our network hardware by which other devices identify it in a network. IP address stands for Internet Protocol address which is a unique number or a numerical representation that uniquely identifies a specific interface on the network. Each device that is connected to internet an IP address is assigned to it for its unique identification.

Addresses in IPv4 are 32-bits long example, 12.244.233.165

And Addresses in IPv6 are 128-bits
example, 2001:0db8:0000:0000:0000:ff00:0042:7879

Subnetting is the practice of dividing a network into two or more smaller networks. It increases routing efficiency, enhances the security of the network and reduces the size of the broadcast domain.

Consider the following example:

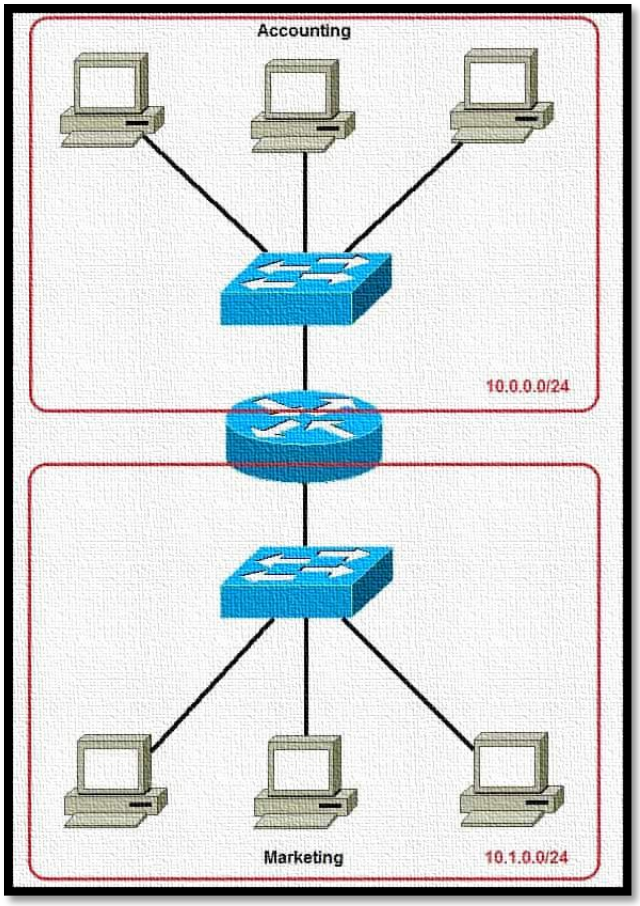


In the picture above we have one huge network: **10.0.0.0/24**. All hosts on the network are in the same subnet, which has the following disadvantages:

- **a single broadcast domain** – all hosts are in the same broadcast domain. A broadcast sent by any device on the network will be processed by all hosts, creating lots of unnecessary traffic.

- **network security** – each device can reach any other device on the network, which can present security problems. For example, a server containing sensitive information shouldn't be in the same network as user's workstations.
- **organizational problems** – in a large networks, different departments are usually grouped into different subnets. For example, you can group all devices from the **Accounting** department in the same subnet and then give access to sensitive financial data only to hosts from that subnet.

The network above could be subnetted like this:



Now, two subnets were created for different departments: **10.0.0.0/24** for Accounting and **10.1.0.0/24** for Marketing. Devices in each subnet are now in a different broadcast domain. This will reduce the amount of traffic flowing on the network and allow us to implement packet filtering on the router.

OR Q.4 (a) Give differences between TCP and UDP. 03

Key	TCP	UDP
Definition	It is a communications protocol, using which the data is transmitted between systems over the network. In this, the data is transmitted in the form of packets. It includes error-checking, guarantees the delivery and preserves the order of the data packets.	It is same as the TCP protocol except this doesn't guarantee the error-checking and data recovery. If you use this protocol, the data will be sent continuously, irrespective of the issues in the receiving end.
Design	TCP is a connection-oriented protocol.	UDP is a connectionless protocol.
Reliability	TCP is more reliable as it provides error checking support and also guarantees delivery of data to the destination router.	UDP, on the other hand, provides only basic error checking support using checksum. So, the delivery of data to the destination cannot be guaranteed in UDP as in case of TCP.
Data transmission	In TCP, the data is transmitted in a particular sequence which means that packets arrive in-order at the receiver.	There is no sequencing of data in UDP in order to implement ordering it has to be managed by the application layer.
Performance	TCP is slower and less efficient in performance as compared to UDP. Also TCP is heavy-weight as compared to UDP.	UDP is faster and more efficient than TCP.
Retransmission	Retransmission of data packets is possible in TCP in case packet get lost or need to resend.	Retransmission of packets is not possible in UDP.

Sequencing	The Transmission Control Protocol has a function that allows data to be sequenced (TCP). This implies that packets arrive at the recipient in the sequence they were sent.	In UDP, there is no data sequencing. The application layer must control the order if it is necessary.
Header size	TCP uses a variable-length (20-60) bytes header.	UDP has a fixed-length header of 8 bytes.
Handshake	Handshakes such as SYN, ACK, and SYNACK are used.	It's a connectionless protocol, which means it doesn't require a handshake.
Broadcasting	Broadcasting is not supported by TCP.	Broadcasting is supported by UDP.
Examples	HTTP, HTTPs, FTP, SMTP, and Telnet use TCP.	DNS, DHCP, TFTP, SNMP, RIP, and VoIP use UDP.

(b) What is socket? Explain its importance at transport layer protocols. 04

A network socket is a software component within a computer network node that acts as an endpoint for delivering and receiving data. An application programming interface (API) for the networking architecture defines the structure and properties of a socket.

The socket provides bidirectional **FIFO** Communication facility over the network. A socket connecting to the network is created at each end of the communication. Each socket has a specific address. This address is composed of an IP address and a port number.

The core communication mechanism used when establishing and managing communication between two devices at the transport layer is called a socket. Basically, any device that wants to establish a transport layer connection to another device must do so via a socket.

(c) Discuss transport layer multiplexing and demultiplexing concepts.

Multiplexing: Multiplexing is the process of collecting the data from multiple application processes of the sender, enveloping that data with headers and sending them as a whole to the intended receiver.

- In Multiplexing at the Transport Layer, the data is collected from various application processes. These segments contain the source port number, destination port number, header files, and data.
- These segments are passed to the Network Layer which adds the source and destination IP address to get the datagram.

Demultiplexing: Delivering the received segments at the receiver side to the correct app layer processes is called demultiplexing.

- The destination host receives the IP datagrams; each datagram has a source IP address and a destination IP address.
- Each datagram carries 1 transport layer segment.
- Each segment has the source and destination port number.
- The destination host uses the IP addresses and port numbers to direct the segment to the appropriate socket.

Q.5 (a) Discuss CSMA/CD Protocol. 03

Carrier Sense Multiple Access with Collision Detection (CSMA/CD) is a network protocol for carrier transmission that operates in the Medium Access Control (MAC) layer. It senses or listens whether the shared channel for transmission is busy or not, and defers transmissions until the channel is free. The collision detection technology detects collisions by sensing transmissions from other stations. On detection of a collision, the station stops transmitting, sends a jam signal, and then waits for a random time interval before retransmission.

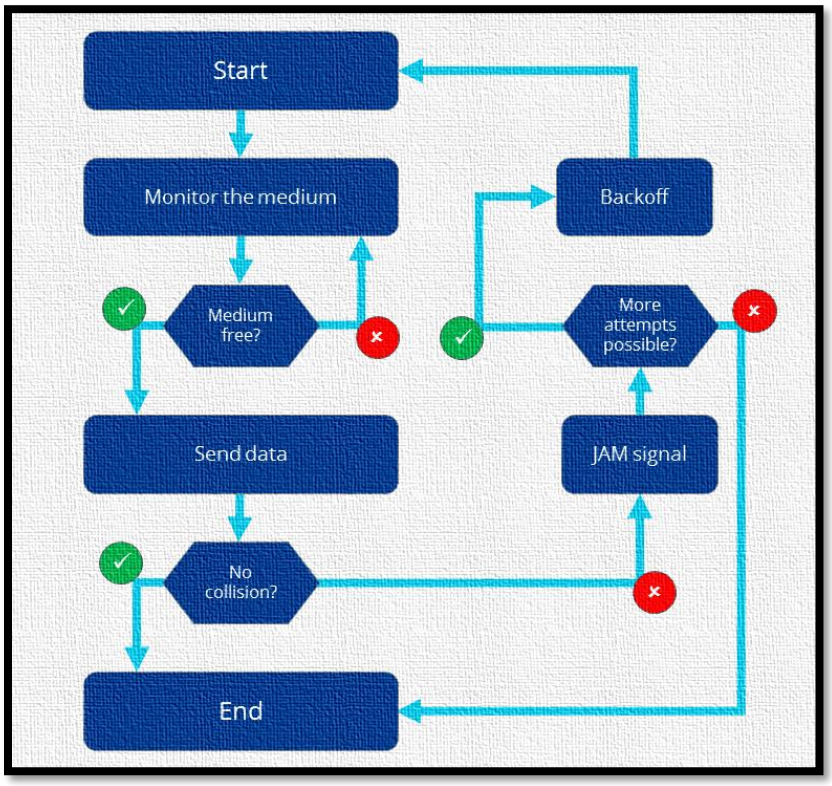
The algorithm of CSMA/CD is:

- When a frame is ready, the transmitting station checks whether the channel is idle or busy.
- If the channel is busy, the station waits until the channel becomes idle.
- If the channel is idle, the station starts transmitting and continually monitors the channel to detect collision.
- If a collision is detected, the station starts the collision resolution algorithm.
- The station resets the retransmission counters and completes frame transmission.

The algorithm of Collision Resolution is:

- The station continues transmission of the current frame for a specified time along with a jam signal, to ensure that all the other stations detect collision.
- The station increments the retransmission counter.

- If the maximum number of retransmission attempts is reached, then the station aborts transmission.
- Otherwise, the station waits for a back off period which is generally a function of the number of collisions and restart main algorithm.



(b) Explain CRC code generation with example. 04

The Cyclic Redundancy Checks (CRC) is the most powerful method for Error-Detection and Correction.

CRC Generator-

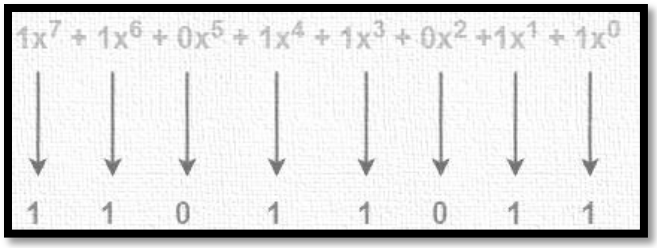
- CRC generator is an algebraic polynomial represented as a bit pattern.
- Bit pattern is obtained from the CRC generator using the following rule-

The power of each term gives the position of the bit and the coefficient gives the value of the bit.

Example-

Consider the CRC generator is $x^7 + x^6 + x^4 + x^3 + x + 1$.

The corresponding binary pattern is obtained as-



Thus, for the given CRC generator, the corresponding binary pattern is 11011011.

(c) Describe Go Back N and Selective Repeat protocol. 07

Both Go-Back-N Protocol and Selective Repeat Protocol are the types of sliding window protocols. The main difference between these two protocols is that after finding the suspect or damage in sent frames go-back-n protocol re-transmits all the frames whereas selective repeat protocol re-transmits only that frame which is damaged.

GO-BACK-N SLIDING WINDOW PROTOCOL	SELECTIVE REPEAT SLIDING WINDOW PROTOCOL
It acts by retransmitting all the frames in the particular order after a damaged or corrupted frame is encountered	It acts by retransmitting only those frames which are damaged or corrupted while transmission
It uses cumulative acknowledgements	It uses independent acknowledgement
In this protocol, there is a great amount of wastage of the bandwidth, if there is a high error rate in the transmission	In this protocol, there is less wastage as compared to Go-Back-N sliding window protocol in retransmission of frames
In this, there is absolutely no need for sorting the sender or the receiver side	In this protocol, there is a need for the sorting of frames that too on the receiver side to maintain the proper sequence of frames
Sliding window size of receiver is 1. For sender, it is N	Sliding window size of receiver is N. For sender, it is same as receiver which is N

It is fairly less complicated than other protocols	It is slightly more complicated solely because it uses strategies, logics, etc. which is not present in all protocols
The receiver does not store any frame that is received after encountering the damaged frame, till the damaged frame is retransmitted	The receiver stores the frames that are received after encountering the damaged frame in a buffer, till the damaged frame is resent by the sender
It is more frequently used as compared to Selective repeat Sliding Window	It is less frequently used as compared to Go-Back-N

OR Q.5 (a) Discuss parity check for error detection in data transfer. 03

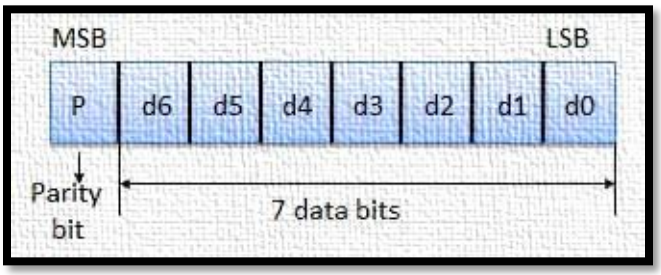
How to Detect and Correct Errors?

To detect and correct the errors, additional bits are added to the data bits at the time of transmission.

- The additional bits are called **parity bits**. They allow detection or correction of the errors.
- The data bits along with the parity bits form a **code word**.

Parity Checking of Error Detection

It is the simplest technique for detecting and correcting errors. The MSB of an 8-bits word is used as the parity bit and the remaining 7 bits are used as data or message bits. The parity of 8-bits transmitted word can be either even parity or odd parity.

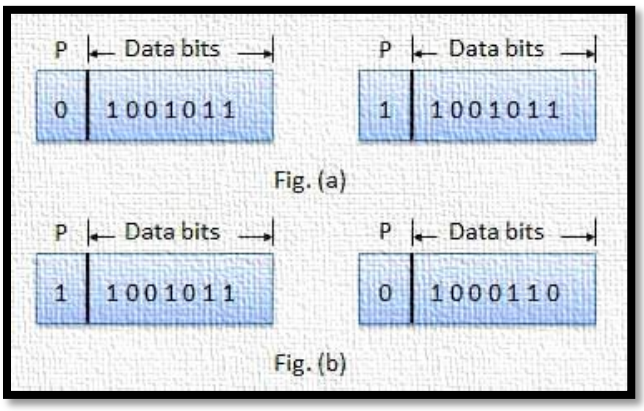


Even parity -- Even parity means the number of 1's in the given word including the parity bit should be even (2,4,6,....).

Odd parity -- Odd parity means the number of 1's in the given word including the parity bit should be odd (1,3,5,....).

Use of Parity Bit

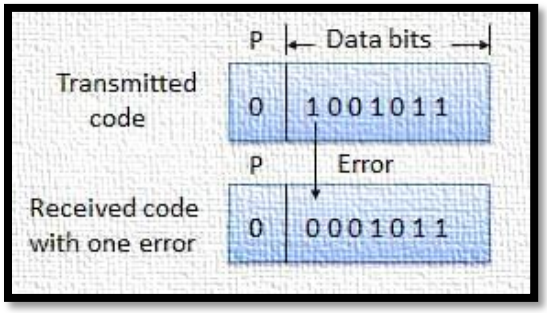
The parity bit can be set to 0 and 1 depending on the type of the parity required.



- For even parity, this bit is set to 1 or 0 such that the no. of "1 bits" in the entire word is even. Shown in fig. (a).
- For odd parity, this bit is set to 1 or 0 such that the no. of "1 bits" in the entire word is odd. Shown in fig. (b).

How Does Error Detection Take Place?

Parity checking at the receiver can detect the presence of an error if the parity of the receiver signal is different from the expected parity. That means, if it is known that the parity of the transmitted signal is always going to be "even" and if the received signal has an odd parity, then the receiver can conclude that the received signal is not correct. If an error is detected, then the receiver will ignore the received byte and request for retransmission of the same byte to the transmitter.



(b) What do you mean by random access protocols? Explain slotted ALOHA in brief. 04

The data link layer is used in a computer network to transmit the data between two devices or nodes. It divides the layer into parts such as **data link control** and the **multiple access resolution/protocol**. The upper layer has the responsibility to flow control and the error control in the data link layer, and hence it is termed as **logical of data link control**. Whereas the lower sub-layer is used to handle and reduce the collision or multiple access on a channel. Hence it is termed as **media access control** or the multiple access resolutions.

Random Access Protocol

In this protocol, all the station has the equal priority to send the data over a channel. In random access protocol, one or more stations cannot depend on another station nor any station control another station. Depending on the channel's state (idle or busy), each station transmits the data frame. However, if more than one station sends the data over a channel, there may be a collision or data conflict. Due to the collision, the data frame packets may be lost or changed. And hence, it does not receive by the receiver end.

Following are the different methods of random-access protocols for broadcasting frames on the channel.

- Aloha
- CSMA
- CSMA/CD
- CSMA/CA

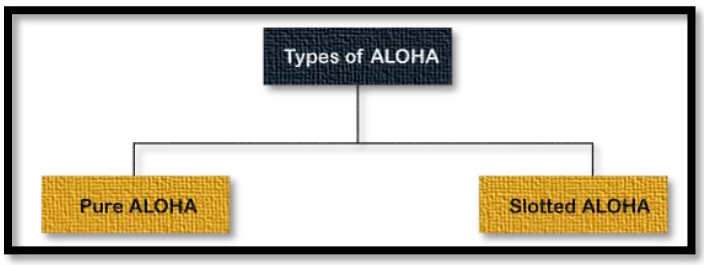
ALOHA Random Access Protocol

It is designed for wireless LAN (Local Area Network) but can also be used in a shared medium to transmit data. Using this method, any station can transmit data across a network simultaneously when a data frameset is available for transmission.

Aloha Rules

1. Any station can transmit data to a channel at any time.
2. It does not require any carrier sensing.
3. Collision and data frames may be lost during the transmission of data through multiple stations.
4. Acknowledgment of the frames exists in Aloha. Hence, there is no collision detection.

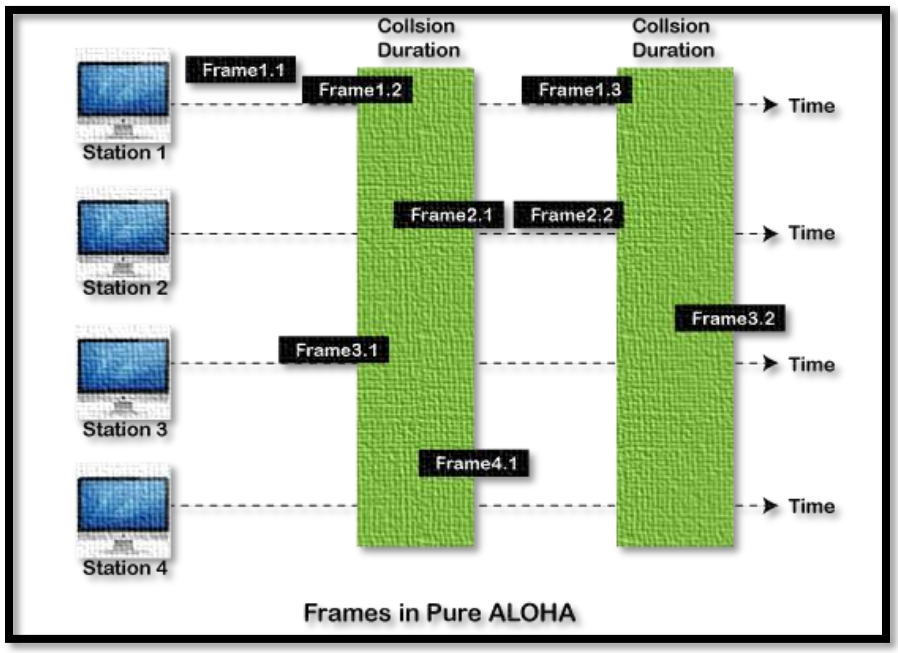
5. It requires retransmission of data after some random amount of time.



Pure Aloha

Whenever data is available for sending over a channel at stations, we use Pure Aloha. In pure Aloha, when each station transmits data to a channel without checking whether the channel is idle or not, the chances of collision may occur, and the data frame can be lost. When any station transmits the data frame to a channel, the pure Aloha waits for the receiver's acknowledgment. If it does not acknowledge the receiver end within the specified time, the station waits for a random amount of time, called the backoff time (Tb). And the station may assume the frame has been lost or destroyed. Therefore, it retransmits the frame until all the data are successfully transmitted to the receiver.

1. The total vulnerable time of pure Aloha is 2 * Tfr.
2. Maximum throughput occurs when G = 1/ 2 that is 18.4%.
3. Successful transmission of data frame is S = G * e ^ - 2 G.

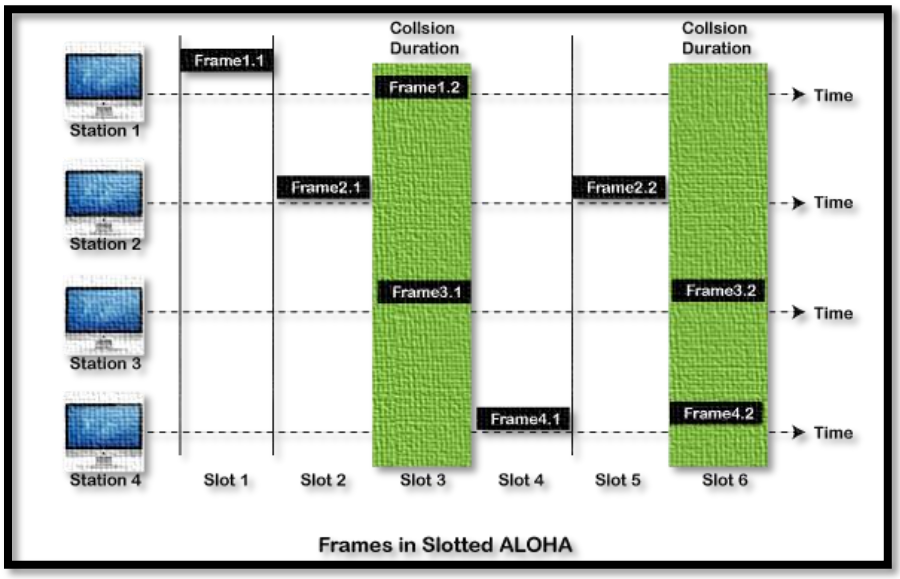


As we can see in the figure above, there are four stations for accessing a shared channel and transmitting data frames. Some frames collide because most stations send their frames at the same time. Only two frames, frame 1.1 and frame 2.2, are successfully transmitted to the receiver end. At the same time, other frames are lost or destroyed. Whenever two frames fall on a shared channel simultaneously, collisions can occur, and both will suffer damage. If the new frame's first bit enters the channel before finishing the last bit of the second frame. Both frames are completely finished, and both stations must retransmit the data frame.

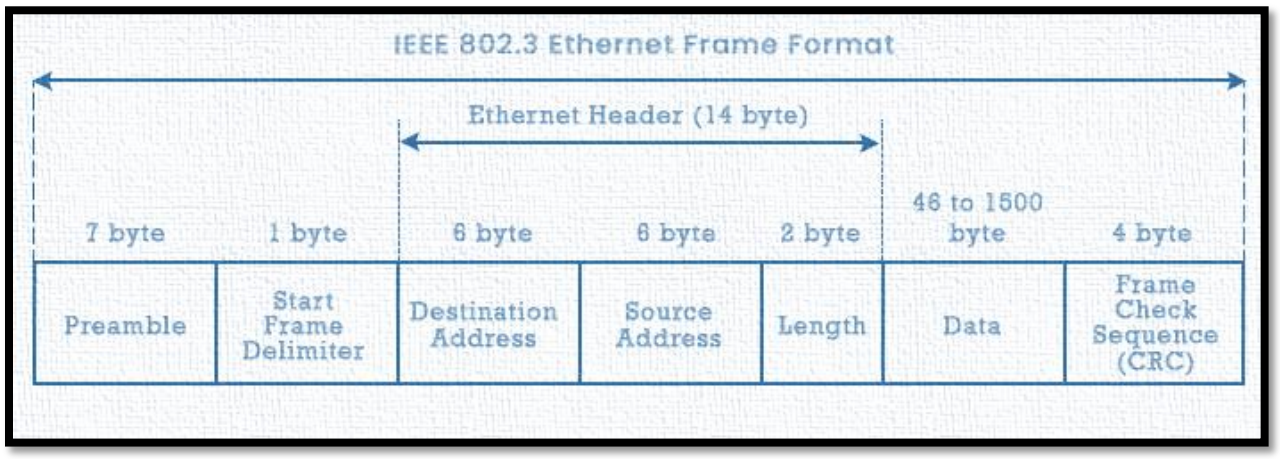
Slotted Aloha

The slotted Aloha is designed to overcome the pure Aloha's efficiency because pure Aloha has a very high possibility of frame hitting. In slotted Aloha, the shared channel is divided into a fixed time interval called **slots**. So that, if a station wants to send a frame to a shared channel, the frame can only be sent at the beginning of the slot, and only one frame is allowed to be sent to each slot. And if the stations are unable to send data to the beginning of the slot, the station will have to wait until the beginning of the slot for the next time. However, the possibility of a collision remains when trying to send a frame at the beginning of two or more station time slot.

1. Maximum throughput occurs in the slotted Aloha when G = 1 that is 37%.
2. The probability of successfully transmitting the data frame in the slotted Aloha is S = G * e ^ - 2 G.
3. The total vulnerable time required in slotted Aloha is Tfr.



(c) Draw and explain Ethernet Frame Structure. 07



1. Preamble-

- It is a 7 byte field that contains a pattern of alternating 0's and 1's.
- It alerts the stations that a frame is going to start.
- It also enables the sender and receiver to establish bit synchronization.

2. Start Frame Delimiter (SFD)-

- It is a 1 byte field which is always set to 10101011.
- The last two bits "11" indicate the end of Start Frame Delimiter and marks the beginning of the frame.

NOTES

- The above two fields are added by the physical layer and represents the physical layer header.
- Sometimes, Start Frame Delimiter (SFD) is considered to be a part of Preamble.
- That is why, at many places, Preamble field length is described as 8 bytes.

3. Destination Address-

- It is a 6 byte field that contains the MAC address of the destination for which the data is destined.

4. Source Address-

- It is a 6 byte field that contains the MAC address of the source which is sending the data.

5. Length-

- It is a 2 byte field which specifies the length (number of bytes) of the data field.
- This field is required because Ethernet uses variable sized frames.

NOTES

- The maximum value that can be accommodated in this field = $2^{16} - 1 = 65535$.
- But it does not mean maximum data that can be sent in one frame is 65535 bytes.
- The maximum amount of data that can be sent in a Ethernet frame is 1500 bytes.
- This is to avoid the monopoly of any single station.

The following three fields collectively represents the **Ethernet Header**–

- Destination Address (6 bytes)
- Source Address (6 bytes)
- Length (2 bytes)

Thus, Ethernet Header Size = 14 bytes.

6. Data-

- It is a variable length field which contains the actual data.
- It is also called as a **payload field**.
- The length of this field lies in the range [46 bytes , 1500 bytes].
- Thus, in a Ethernet frame, minimum data has to be 46 bytes and maximum data can be 1500 bytes.

Minimum Length of Data Field

- Ethernet uses CSMA / CD as access control method to deal with collisions.
- For detecting the collisions, CSMA / CD requires-

Minimum length of data packet = 2 x Propagation delay x Bandwidth

- Substituting the standard values of Ethernet, it is found that minimum length of the Ethernet frame has to be 64 bytes starting from the destination address field to the CRC field and 72 bytes including the Preamble and SFD fields.
- Therefore, minimum length of the data field has to be = 64 bytes – (6+6+2+4) bytes = 46 bytes

Maximum Length of Data Field

- The maximum amount of data that can be sent in a Ethernet frame is 1500 bytes.
- This is to avoid the monopoly of any single station.
- If Ethernet allows the frames of big sizes, then other stations may not get the fair chance to send their data.

7. Frame Check Sequence (CRC)-

- It is a 4 byte field that contains the CRC code for error detection.

Thank You